Warm Up: What is the syntax for slicing values from a list?

Vectors and Kinematics: 2-D Applications

SciTeens SOLID Physics Day 3

Recap

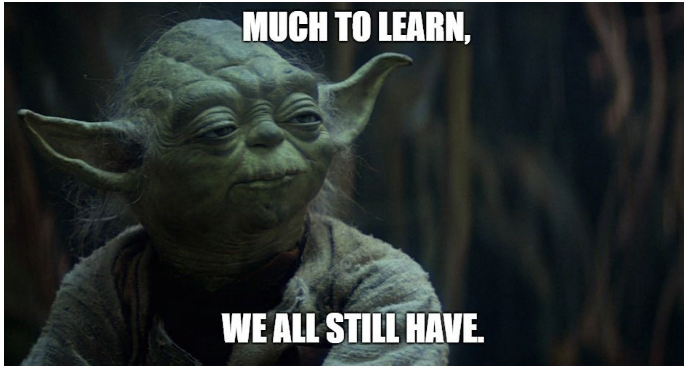
- So far, we’ve covered defining vectors in general and the specific vectors involved in particle motion, along with understanding the kinematic equations and the variables involved in them
- Now, we can combine a few of these concepts -- vectors, the kinematic equations, and basic trigonometry -- to use the kinematic equations in 2 (or more!) dimensions

Learning Objectives

In this lesson, we will combine the two main topics we have covered so far -- vectors and 1D kinematics -- along with some smaller topics we will cover along the way, to eventually master 2D kinematics

Here are some things we need to cover before this mastery:

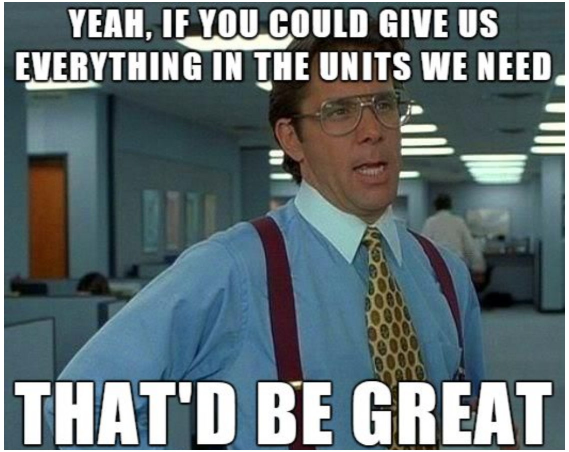
- Dimensional analysis and converting units
- Angle of elevation and forming right triangles based off of them
- Keeping track of variables in multiple dimensions and combining them at the end of the problem



Units: To Convert or not to Convert?

Most kinematic calculations are done in m (meters) and s (seconds), so when we are given our values in units we aren't used to, we have 2 choices:

(1) Convert the given units to m and s, and then perform the calculations. Values at the end of the problem can be reported with the old or new units
(2) Use the units given, with the answer having a discrepancy from most other problems



Dimensional Analysis

Dimensional analysis is an intuitive way of converting units using ratios given. The setup and steps will be shown in the examples below:

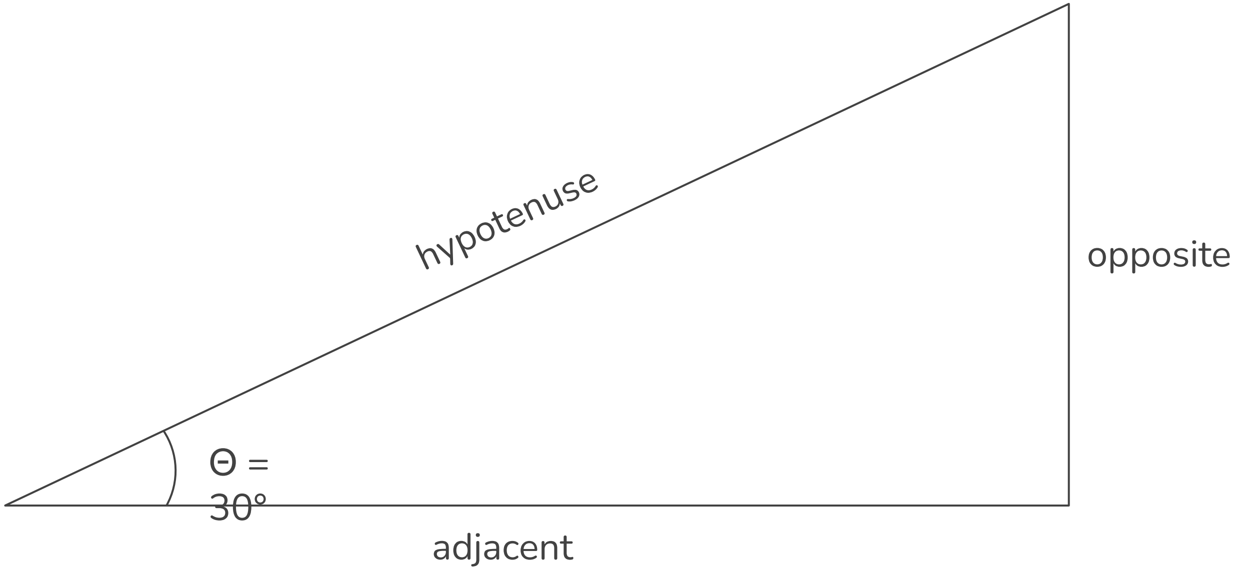
Ex 1: Convert the value 100 km/h to m/s

$$\frac{100\ \text{km}}{1\ \text{h}} \times \frac{1\ \text{h}}{60\ \text{min}} \times \frac{1\ \text{min}}{60\ \text{s}} \times \frac{1000\ \text{m}}{1\ \text{km}} = 27.8\ \text{m/s}$$

Ex 2: Convert the value 50 mph to km/h to m/s

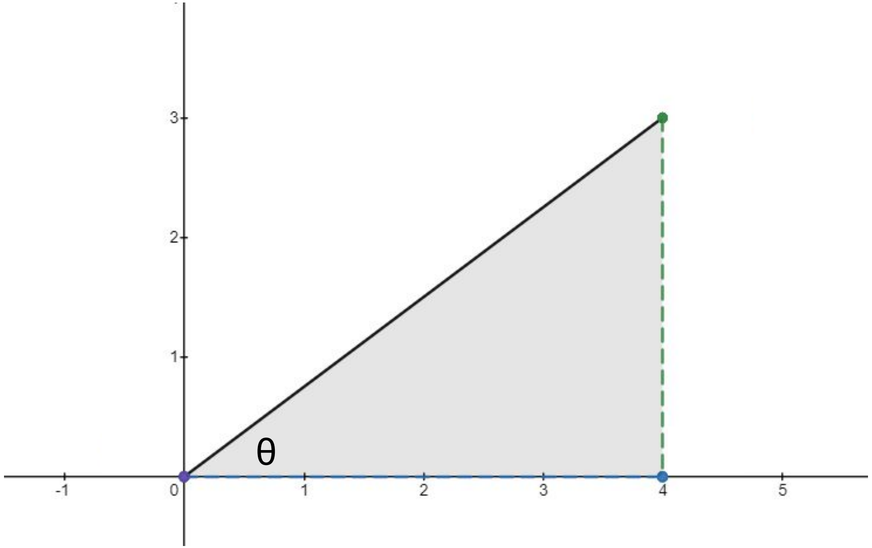
$$\frac{50\ \text{miles}}{1\ \text{h}} \times \frac{1.609\ \text{km}}{1\ \text{mile}} \times \frac{1\ \text{h}}{60\ \text{min}} \times \frac{1\ \text{min}}{60\ \text{s}} \times \frac{1000\ \text{m}}{1\ \text{km}} = 22.3\ \text{m/s}$$

Constructing Triangles Using Angle of Elevation



Motion Vector Components

Suppose a velocity vector (illustrated below) has magnitude v = 5m/s and angle θ = 36.87°. Calculate the x and y vector components and interpret them in the context of the entire vector.



Applying Motion Vector Components to Kinematics

Using the previous example and the vector components derived, suppose we are also given that the acceleration of the particle in the x direction (a_x) is 2m/s/s and $a_y = 0$. We can also assume that the particle starts at position x = y = 0. After 5 seconds of traveling, what is the particle's final x position, y position, and net displacement (this involves recombining the vector components!)?

Practice Problems

[Practice Link]

Additional Practice/Resources

Resource #1

Resource #2

Resource #3

DO THIS!!!

Additional Practice #1

Additional Practice #2